

RECOIL-OPERATED MACHINE-GUNS (1884–95)

In 1884, Sir Hiram Maxim invented a machine-gun (see pp. 184–85) that worked by recoil-action instead of being hand-cranked like earlier machine-guns (see pp. 136–39). Maxim had initially tried to harness the energy of a gun's recoil to automate its action in rifles such as the Peabody-Martini and Winchester. He ultimately perfected the process in his machine-gun. The Maxim machine-gun was fully automatic, meaning it could fire continuously while the trigger was kept pulled. Within 10 years, armies in the UK, Germany, and Russia had adopted this weapon.

▼ MAXIM EARLY PATTERN MACHINE-GUN

Date 1885

Origin Germany/UK

Barrel 28in (72cm)

Caliber .45in

Seen here is an 1885 model of Maxim's first machine-gun. At first, orders for the Maxim gun were hampered by the fact that it produced clouds of gunpowder smoke, but once it was allied to smokeless powder (see pp. 142–43), it became a truly significant battlefield weapon. Like many early machine-guns, this one depended on a water-cooling system to temper heat generated during continuous firing in order to extend its barrel life.

Rear sight (unfolded)

Trigger

Elevation adjustment

Tripod leg

FULL VIEW

Water coolant jacket

Water coolant jacket

Elevation wheel

Gun on anti-aircraft mounting

Naval pintle mounting, modified during World War I for a ship's defense against attacking aircraft

Recoil booster enhances recoil to aid reloading

► MAXIM 1-POUNDER "POM-POM"

Date 1890

Origin UK

Barrel 43in (109cm)

Caliber 37mm

The "Pom-Pom"—so-called because of the noise it made when in use—was an enlarged version of Maxim's .45in Gardner-Gatling caliber machine-gun (right). It was the world's first autocannon—unlike a machine-gun, it fired shells rather than bullets. The "Pom-Pom" served as an artillery weapon and an anti-aircraft gun in World War I.

Foresight

Water coolant jacket

